Week 3: GenAI - LLM APIs in Google Sheets

Introduction

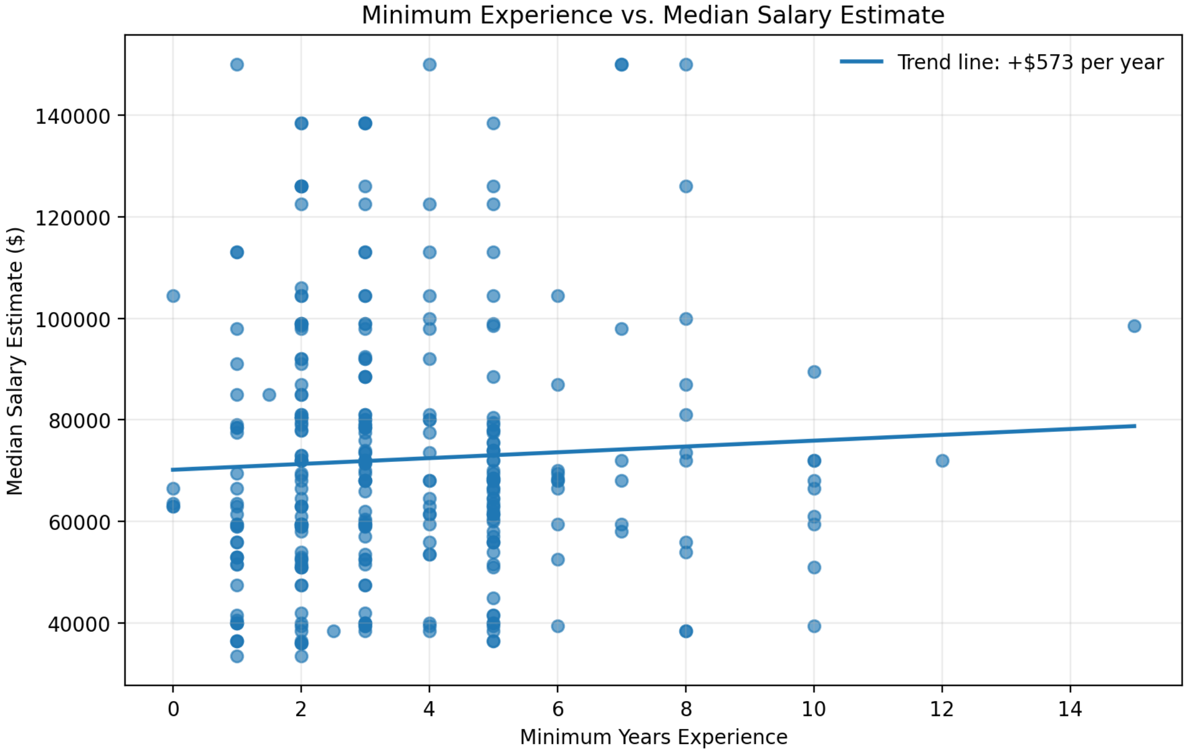
This dataset contains a sample of **Data Analyst job postings**, with extracted information about salary, experience requirements, and preferred programming or data-analysis languages. The key variables used in the visuals are **Median Salary Estimate**, which represents the estimated midpoint salary for each posting; **Minimum Years Experience**, which captures the minimum experience requested by the employer; and **Data Languages**, which categorizes postings based on whether they mention **R**, **Python**, **Both**, or **Neither**.

Because some experience values may contain missing or non-numeric text, the numeric fields were cleaned before plotting. Rows with invalid experience or salary values were excluded from the scatter plot, while the box plot groups available salary estimates by language category.

Key inspection results:

- Dataset size: **400 rows × 13 original columns**
- Required columns were present:
  - Median Salary Estimate
  - Minimum Years Experience
  - Data Languages
- Minimum Years Experience: **310 usable numeric values**, **90 missing/non-numeric**
- Standardized Data Languages counts:
  - R: **15**
  - Python: **62**
  - Both: **68**
  - Neither: **255**

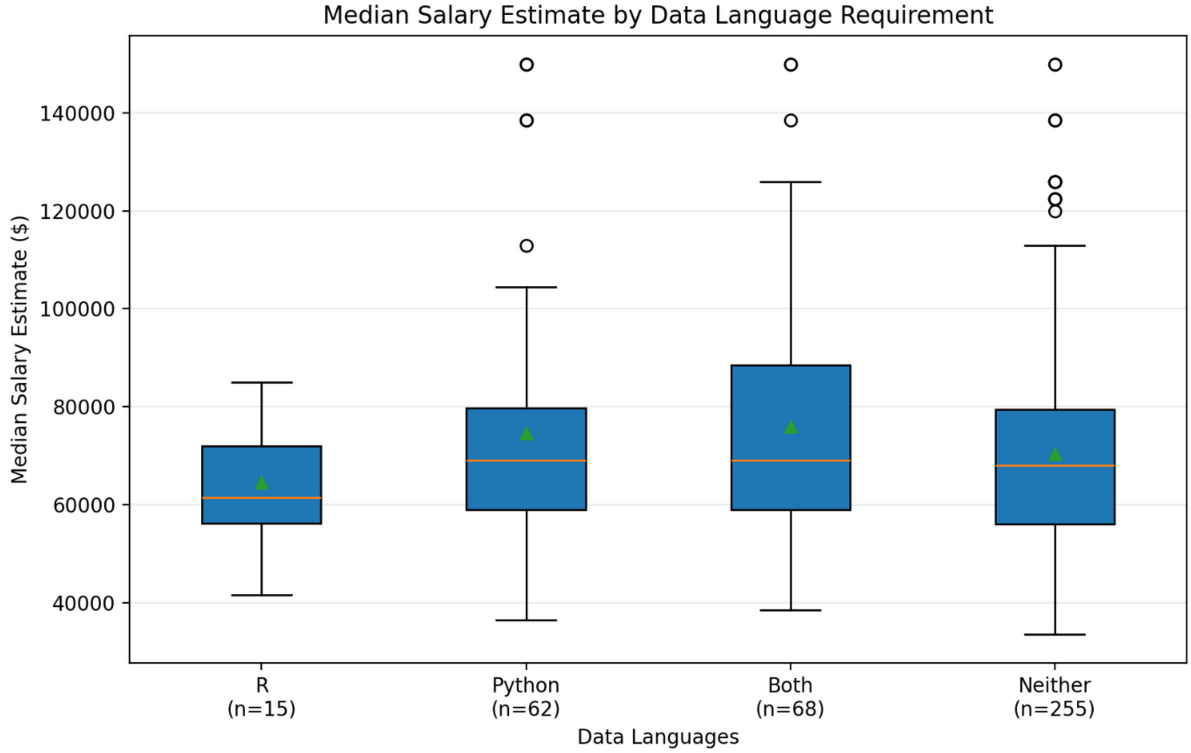
Years of Experience and Salary Estimate



The scatter plot compares **Minimum Years Experience** on the x-axis with **Median Salary Estimate** on the y-axis. Overall, the data shows a wide spread of salaries across most experience levels, especially between about **2 and 5 years of experience**, where many postings are concentrated.

The scatter plot shows only a **very weak positive relationship** between required experience and salary. The trend line suggests about **$573 more per additional year of minimum required experience**, but the correlation is very low, so experience alone does not explain salary differences well in this sample.

Salary Estimate and Data Languages



The box plot compares **Median Salary Estimate** across the four **Data Languages** categories: **R**, **Python**, **Both**, and **Neither**.

The postings that mention **Python**, **Both Python and R**, or **Neither** generally show higher salary ranges than postings mentioning only **R**. The **Both** category appears to have a relatively high average salary and a wider spread, suggesting that jobs asking for multiple language skills may include more varied or advanced roles. The **Python** and **Neither** categories also include several high-salary outliers, indicating that some roles in these groups offer substantially above-typical compensation.

Overall, the box plot suggests that language requirements may be associated with salary differences, but the overlap between groups means language category alone does not fully explain salary variation.

Reflections

Overall, this process was fairly simple to follow. There are a lot of steps to use LLM APIs but it is good to see that big companies are already trying to incorporate their models in their applications i.e. Gemini in Google Sheets. Apart from learning to incorporate LLMs in Google sheets, the application of this learning is immense and I find it very useful in my work as a data analyst.